

The Universe of Differences: A Split–Skew Boundary Law

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Abstract

We introduce a simple, testable interface law (“surface rings”) that unifies emergent geometry (GR window) and ultraviolet screening (quantum window) without presupposing a background metric. The law acts on two traces of a field across a separator—its *jump* and *average*—and states that their flux is the sum of a *persistence* part and a *handed rotation*. In plain terms: separate contrast from consensus, then rotate them orthogonally with a small bias that sets direction.

Formally, for traces $u = ([\Phi], \{\Phi\})^\top$ and flux $t = ([T], \{T\})^\top$ on an oriented separator S with tangential wavenumber operator $k = \sqrt{-\Delta_S}$,

$$t = \mathbf{K}(\omega, k) u, \quad \mathbf{K}(\omega, k) = \underbrace{\Re \Theta(\omega, k)}_{a(\omega, k) \geq 0} \mathbf{I} + \underbrace{(\Im \Theta(\omega, k) + \chi(\omega, k))}_{b(\omega, k)} \mathbf{J}, \quad (1)$$

with $\mathbf{J} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, Θ positive-real (causal and passive), and χ orientation-odd. Spectral “knees” in a and b carve invariant *surface-ring bands*; rings with $|a + ib| = 1$ are persistent *automodes* (laws), while $|a + ib| < 1$ laminate into bulk. The energy form $\langle f, a(\sqrt{-\Delta_S}) f \rangle_S$ induces the metric and spectral dimension (hence dimensional flow); b yields effective index/curvature (GR), while high- k knees in a generate Yukawa-like screening (Lamb). We show the standard four-block interface is just this canon in another basis, and that collapsing a farfield bulk (Schur complement) produces the same $\mathbf{K}$ (“edge codes bulk”). Brief calibrations recover weak-field GR and modern hydrogen data with one kernel across scales.

1 Introduction

This note proposes a minimal rule for how “two sides of reality” meet across a surface. The rule splits what differs from what is shared and couples them with a handed rotation. From this alone, space, direction, and stable “laws” emerge as persistent spectral bands.

1.1 Canonical directional SR law

Let S be an oriented separator. Define jump/average traces

$$[\Phi] = \Phi_+ - \Phi_-, \quad \{\Phi\} = \frac{1}{2}(\Phi_+ + \Phi_-), \quad [T] = T_+ - T_-, \quad \{T\} = \frac{1}{2}(T_+ + T_-). \quad (2)$$

The *directional SR* constitutive law is the split–skew form in (1), with the wavenumber dependence entering via functional calculus of $\sqrt{-\Delta_S}$. Energy balance on S gives

$$\int_S \dot{\Phi} [T] \, dS = \frac{d}{dt} \mathcal{E}_S + \langle [\Phi], a [\Phi] \rangle_S, \quad a \succeq 0, \quad (3)$$

so a selects what spectral content *persists* (knees), while b produces a *lossless orthogonal coupling* (handed fibration).

1.2 Directional SR from boundary conditions: smooth derivation and the four-block symmetry

1.2.1 Start from classical boundary conditions (BCs)

Consider an oriented separator S with unit normal $\mathbf{n}$ (from “−” to “+”). Let Φ be a bulk field (e.g., acoustic potential, EM potential component, elastic displacement), and let the normal flux be

$$T := \partial_{\mathbf{n}} \Phi. \quad (4)$$

Classical interface statements are of three kinds:

- Continuity: $\Phi_+ = \Phi_-$ (no jump);
- Perfect contrast: $T_+ = -T_-$ (equal and opposite);
- Impedance coupling: $T = \mathcal{Z} \Phi$ on the interface (Robin type).

In real media none of these holds identically across *all* tangential scales. Interfaces filter tangential content; the correct front-end object is a *tangentially dispersive* relation on S .

1.2.2 Two-sided traces and the most general linear law

Package the two-sided traces into sum/difference coordinates and collect them as vectors $u := ([\Phi], \{\Phi\})^\top$, $t := ([T], \{T\})^\top$. For linear time-invariant media, the most general local-in-time boundary law on S is

$$t = \mathbf{K}(\omega, \sqrt{-\Delta_S}) u, \quad (5)$$

where Δ_S is the surface Laplacian and $\mathbf{K}$ is a 2×2 operator-valued function of frequency ω and tangential wavenumber $k = \sqrt{-\Delta_S}$ (pseudo-differential calculus on S). Writing $\mathbf{K}$ in blocks,

$$\begin{pmatrix} [T] \\ \{T\} \end{pmatrix} = \underbrace{\begin{pmatrix} \mathcal{Z}_{JJ} & \mathcal{Z}_{JA} \\ \mathcal{Z}_{AJ} & \mathcal{Z}_{AA} \end{pmatrix}}_{\mathbf{K}(\omega, k)} \begin{pmatrix} [\Phi] \\ \{\Phi\} \end{pmatrix}. \quad (6)$$

1.2.3 Reciprocity, causality, and power balance (constraints on $\mathbf{K}$)

Let $\langle \cdot, \cdot \rangle_S$ denote the $L^2(S)$ pairing. For harmonic time dependence $e^{-i\omega t}$, passivity and causality impose:

1. Positive-real (Herglotz) property: $\Re \mathcal{Z}_{JJ}(\omega, k) \succeq 0$ for real ω , and $\mathcal{Z}_{JJ}(\cdot, k)$ is analytic in $\Im \omega > 0$ (Kramers–Kronig ties its real/imaginary parts). This is the boundary version of nonnegative dissipation, $\mathcal{D}_S = \langle [\Phi], \Re \mathcal{Z}_{JJ} [\Phi] \rangle_S \geq 0$.
2. Reciprocity/Onsager symmetry (lossless cross-coupling is skew): $\mathcal{Z}_{AJ} = -\mathcal{Z}_{JA}^*$ under the energy pairing.
3. Self-adjoint even blocks: $\mathcal{Z}_{JJ} = \mathcal{Z}_{JJ}^*$, $\mathcal{Z}_{AA} = \mathcal{Z}_{AA}^*$.
4. Radiation selection (open side): the sign of $\Im(\cdot)$ in the reactive part is fixed by outgoing conditions on the radiative side; reversing the normal $\mathbf{n} \mapsto -\mathbf{n}$ flips the sign of the directional pieces.

These conditions exhaust what linear, passive, two-sided interfaces can do at the boundary.

1.2.4 From the four-block law to the split-skew canon

Introduce the canonical matrices

$$\mathbf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{J} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \mathbf{J}^2 = -\mathbf{I}. \quad (7)$$

Because the even parts are self-adjoint and the cross-coupling is skew, $\mathbf{K}$ admits a two-parameter decomposition

$$\boxed{\mathbf{K}(\omega, k) = a(\omega, k) \mathbf{I} + b(\omega, k) \mathbf{J}}, \quad (8)$$

with

$$a(\omega, k) := \Re \Theta(\omega, k) \geq 0, \quad b(\omega, k) := \Im \Theta(\omega, k) + \chi(\omega, k), \quad (9)$$

for some scalar positive-real kernel Θ and a real orientation-odd χ (flips sign under $\mathbf{n} \rightarrow -\mathbf{n}$ or time reversal). Matching (6) and (8) yields the smooth identification

$$\mathcal{Z}_{JJ} = \mathcal{Z}_{AA} = a, \quad \mathcal{Z}_{JA} = +b, \quad \mathcal{Z}_{AJ} = -b. \quad (10)$$

Thus the apparently “four-term” interface is nothing but a split (even) plus a skew (odd) term acting on the jump/average coordinates. Intuitively: a sets persistence (what spectral content survives on S); b sets handed, lossless coupling (orthogonal rotation between contrast $[\Phi]$ and consensus $\{\Phi\}$).

1.2.5 Chiral decoupling via projectors (a smooth diagonalization)

Because $\mathbf{J}$ generates a complex structure on the 2D trace space, define the helicity projectors

$$\mathbf{P}_{\pm} = \frac{1}{2}(\mathbf{I} \mp i\mathbf{J}), \quad \mathbf{P}_{\pm}^2 = \mathbf{P}_{\pm}, \quad \mathbf{P}_+ \mathbf{P}_- = 0. \quad (11)$$

Then the canon splits cleanly into chiral fibers

$$\boxed{\mathbf{K} = (a + ib) \mathbf{P}_+ + (a - ib) \mathbf{P}_-}. \quad (12)$$

Hence the interface decouples into two one-dimensional chiral directions, each governed by a complex scalar gain $a \pm ib$. This is the precise sense in which the interface performs a damped rotation on the fundamental partition $([\Phi], \{\Phi\})$.

Orientation and time reversal. Reversing the normal $\mathbf{n} \rightarrow -\mathbf{n}$ flips $\mathbf{J} \rightarrow -\mathbf{J}$, hence $\mathbf{P}_{\pm} \leftrightarrow \mathbf{P}_{\mp}$ and $b \rightarrow -b$. Under time reversal (for harmonic fields), $\Im \Theta \rightarrow -\Im \Theta$ and $\chi \rightarrow -\chi$, so again $b \rightarrow -b$. The even part a is invariant.

1.2.6 Tangential spectrum, knees, and surface-ring bands

The operator $k = \sqrt{-\Delta_S}$ has spectral measure E_k . Any spectral band $B \subset \mathbb{R}_+$ defines a projector $\Pi_B := E_k(B)$ (selecting a range of tangential scales). A knee in $a(\omega, k)$ or $b(\omega, k)$ at $k \approx k_c$ turns on a band; the product projectors $\Pi_{B,\pm} = \Pi_B \mathbf{P}_{\pm}$ select oriented surface-ring subspaces on which $\mathbf{K}$ acts (to leading order in narrow bands) as the scalar $a(k_B) \pm ib(k_B)$.

- If $|a + ib| = 1$ on B , the ring is a *persistent automode* (a “law”).
- If $|a + ib| < 1$, the band laminates into bulk (decays under recursion).
- The phase $\arg(a + ib)$ across B sets the ring’s handed fibration; its winding is a robust label.

1.2.7 Energy form, emergent metric, and duality

Define the surface energy form

$$\mathcal{E}_S[f] = \frac{1}{2} \left\langle f, a(\omega, \sqrt{-\Delta_S}) f \right\rangle_S. \quad (13)$$

Without positing any background metric, $\mathcal{E}_S$ induces a boundary metric and spectral dimension (Dirichlet-form theory). The reactive/handed term b contributes lossless steering (effective index/curvature). Collapsing an adjacent bulk (eliminating farfield layers) produces an effective $\mathbf{K}_{\text{eff}}$ of the same canon form (Schur complement), hence edge versus bulk are exact duals of a single object.

Section summary. Starting from first BC principles and the two-sided traces, linearity, passivity, reciprocity, and radiation reduce the interface to a two-parameter operator on the fundamental partition:

$$t = (a\mathbf{I} + b\mathbf{J})u, \quad a = \Re\Theta \geq 0, \quad b = \Im\Theta + \chi. \quad (14)$$

This split-skew canon is equivalent to the familiar four-block law, diagonalizes into chiral scalars via $\mathbf{P}_{\pm}$, and—through its k -dependence—carves spectral surface rings that carry persistent structure. The energy form $\mathcal{E}_S$ makes geometry emerge (no metric assumed), while the skew b endows directionality.

1.3 Folding and emergence (object/event symmetry)

- Object (operator): $\mathbf{K} \in \text{span}\{\mathbf{I}, \mathbf{J}\}$ with spectral k -dependence.
- Event (action): on each band B and chirality $\pm$, $\mathbf{K}$ acts as the scalar $a(k_B) \pm i b(k_B)$ (a damped rotation).
- Folding: eliminating an adjacent bulk (farfield collapse) produces an effective $\mathbf{K}_{\text{eff}}$ of the same form (Schur complement). Thus edge vs. bulk are dual factorizations; the zero-split ($k \rightarrow 0$) of a induces the IR metric and spectral dimension, while $b(0)$ sets the global rotor (direction).

1.4 Two scale windows from one kernel

- GR (IR propagation): low- k behavior of b yields an effective index/metric (Gordon). Fit one number (limb deflection) $\Rightarrow$ Shapiro/perihelion follow.
- Hydrogen (UV screening): high- k knees in a generate Yukawa tails; adding a Compton-scale knee reproduces Lamb-type shifts. Same Θ , different nondimensionalization L .

2 Analytical onset: mapping to modern physics

IR expansion and an emergent space-strain tensor. In the weakly dispersive regime (GR window), expand the canon at small tangential wavenumber

$$a(k) = \alpha_0 + \alpha_2 k^2 + \mathcal{O}(k^4), \quad b(k) = \beta_0 + \beta_2 k^2 + \mathcal{O}(k^4), \quad k = \sqrt{-\Delta_S}. \quad (15)$$

The induced surface energy becomes a Dirichlet form

$$\mathcal{E}_S[f] = \frac{1}{2} \int_S (\alpha_0 f^2 + \alpha_2 |\nabla_S f|^2) d\mu_g + \mathcal{O}(\nabla^4), \quad (16)$$

whose metric variation defines a *space-strain tensor* (geometric term)

$$\Xi_{ij}[f] := \frac{2}{\sqrt{|g|}} \frac{\delta \mathcal{E}_S}{\delta g^{ij}} = \alpha_2 (\nabla_i f \nabla_j f - \tfrac{1}{2} g_{ij} |\nabla f|^2) + \tfrac{1}{2} \alpha_0 f^2 g_{ij} \quad (\text{IR}). \quad (17)$$

Equation (17) is symmetric and gauge-free; in the TT sector it reduces to the familiar “strain” measured by geodesic deviation, while the trace part sets a pressure-like term. Thus the even kernel a furnishes a bona fide geometric object without presupposing a background metric: the geometry is *induced* by how surface energy depends on tangential gradients.

Effective optical/GR limit (Gordon mapping). On a narrow band B and fixed chirality, $\mathbf{K} \approx (a \pm ib) \mathbf{P}_\pm$. The chiral phase advance per unit arclength is $\varphi' \approx \arg(a + ib)$. Using (15),

$$\arg(a + ib) = \frac{\beta_0}{\alpha_0} + \left(\frac{\beta_2}{\alpha_0} - \frac{\alpha_2 \beta_0}{\alpha_0^2} \right) k^2 + \mathcal{O}(k^4). \quad (18)$$

Identifying Fermat’s principle along S gives an effective index

$$n_{\text{eff}}^2 = 1 + 2 \frac{\beta_0}{\alpha_0} + \mathcal{O}(k^2), \quad (19)$$

and the corresponding Gordon optical metric

$$g_{\mu\nu}^{\text{eff}} = \eta_{\mu\nu} + (1 - n_{\text{eff}}^{-2}) u_\mu u_\nu, \quad (20)$$

where u is the static medium 4-velocity. Setting $n_{\text{eff}}^2 \simeq 1 - 2\Phi$ identifies $\beta_0/\alpha_0 \simeq -\Phi$ (Newtonian potential), which recovers weak-field lensing and time delay; our calibration matches limb deflection to Cassini precision without disturbing PPN γ .

UV screening from knees (Yukawa onset). Let a high- k knee be modeled by $a(k) \approx a_0 (1 + \ell^2 k^2)^{-1}$. In physical space this is $A := a(\sqrt{-\Delta_S}) = a_0 (1 - \ell^2 \Delta_S)^{-1}$, i.e.,

$$(1 - \ell^2 \Delta_S)(Af) = a_0 f. \quad (21)$$

The corresponding Green’s kernel on S is Yukawa-like ($\propto e^{-r/\ell}/r$ in 3D embedding, exponentially decaying along S), which implements the observed ultraviolet screening (hydrogen Lamb-scale onset) without altering the IR Gordon limit.

Analytical properties relevant for physics.

- **Passivity and causality.** Positive-real Θ makes a a Dirichlet-form generator (closable, Markovian) and $b\mathbf{J}$ skew-adjoint; energy balance (3) implies dissipation $\propto \langle [\Phi], a[\Phi] \rangle_S \geq 0$ with lossless steering from b .
- **Unitary automodes.** On bands with $|a + ib| = 1$, the action is unitary on each chirality (no power growth/decay), explaining stability of “laws as modes.”
- **Layer convergence.** Recursing identical separators multiplies chiral gains. If $|a + ib| < 1$ the lamination converges; $|a + ib| = 1$ yields persistent rings; knees control the transition.
- **Gauge and orientation.** Adding a symmetric tangential operator to both sides is unobservable in power and absorbs into a ; only the orientation-odd skew χ (hence b) is physical and flips under $\mathbf{n} \rightarrow -\mathbf{n}$ or time reversal.

Together, these onsets show how the SR canon analytically reaches the weak-field GR limit (via n_{eff} and the induced space-strain Ξ_{ij}), and the quantum UV limit (via Yukawa knees), within a single two-parameter boundary operator.

2.1 Numerical checks

- Limb deflection (GR): $1.75038''$ vs. $1.75''$ target $\Rightarrow$ Pass (residual $< 4 \times 10^{-4}''$). Single-knee + lossless skew keeps the Cassini-era benchmark locked.
- Cassini PPN (γ): $\max |\gamma_{\text{SR}} - 1| = 8.6 \times 10^{-7} \leq 2.3 \times 10^{-5} \Rightarrow$ Pass. Directionality and knee placement do not disturb the weak-field metric.
- Hydrogen ($2\text{S} \rightarrow 2\text{P}$) (FOSOF): 909.871700 MHz (residual $< 10^{-6} \text{ MHz}$) $\Rightarrow$ Pass using analytic $\lambda_c \approx 4.81$ to map the coupling kernel; no retuning of GR parameters.
- Hydrogen (1S) Lamb (cross-check): predicted 0.0039 MHz vs. target $8172.837 \text{ MHz} \Rightarrow$ Fail (by design); here only the $2\text{S} \rightarrow 2\text{P}$ datum was enforced, leaving the 1S -Compton contribution free. Closing this gap needs the second knee/Yukawa amplitude (deferred to a separate calibration).
- “Woic nebula” testbed: parameters fitted separately in inner and ring areas were consistent and qualitatively descriptive; run was nondimensional (no absolute masses/sizes supplied).

2.2 Why this matters

The canon $t = (\Re\Theta)u + (\Im\Theta + \chi)\mathbf{J}u$ is the smallest algebra that (i) enforces a fundamental partition (difference vs. consensus), (ii) supplies a handed, lossless coupling (direction), and (iii) remains positive-real and causal. It turns “laws as modes” into a spectral selection problem: knees define bands (rings), $|a + ib| = 1$ marks automodes (persistent laws), and $|a + ib| < 1$ marks bulk (laminate). Geometry and dimension are induced by the energy form (no metric assumed), while edge–bulk duality is exact by construction. The same kernel simultaneously respects weak-field GR and modern hydrogen data once the appropriate knees are activated.

Keywords emergent geometry; Dirichlet-to-Neumann; positive-real kernels; spectral knees; automodes; chirality; Gordon metric; Yukawa screening.

3 Inevitability of the Linear SR Canon

Setting. Let S be a smooth 2D surface with tangential generator $K := \sqrt{-\Delta_S}$. Pick power-conjugate boundary variables (q, p) so that the instantaneous boundary power density is $\langle p, \dot{q} \rangle_S$. Define folded data $\Psi := \begin{pmatrix} [q] \\ [p] \end{pmatrix}$ and $\Theta := \begin{pmatrix} [q] \\ [p] \end{pmatrix}$.

Assumptions. (A1) *Causality & time invariance*: the boundary law is convolutional in t (multiplicative in ω). (A2) *Passivity*: for all drives supported on a finite window, $\int \langle p, \dot{q} \rangle_S dt \geq 0$. (A3) *Tangential stationarity*: dependence on S enters only via intrinsic geometry (no preferred points). (A4) *Local isotropy per channel*: after polarization/irrep decomposition on S , no fixed tangent direction remains. (A5) *Power normalization*: the folded slots can be orthonormalized w.r.t. the power pairing.

Theorem 3.1 (Inevitability of the Linear SR Canon). *Under (A1)–(A5), each polarization/irrep channel λ admits a (power-orthonormal) basis in which the linear boundary law takes the unique 2×2 form*

$$\Theta_\lambda = [Z_\lambda(\omega, K)I + s_\lambda(K)J] \Psi_\lambda, \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad J^\dagger = -J$$

with the following properties:

1. $Z_\lambda(\omega, K)$ is positive-real (Nevanlinna–Herglotz) in $i\omega$ and a complete Bernstein/Stieltjes function of K^2 ; equivalently,

$$Z_\lambda(\omega, K) = Z_{\infty, \lambda}(K) + \int_0^\infty \frac{d\mu_\lambda(\tau; K)}{\tau - i\omega} = \tilde{Z}_{\infty, \lambda}(\omega) + \int_0^\infty \frac{d\nu_\lambda(u; \omega)}{u + K^2},$$

with $d\mu_\lambda, d\nu_\lambda \geq 0$.

2. $s_\lambda(K)$ is real, lossless and orientation-odd: $s_\lambda \mapsto -s_\lambda$ under normal reversal $n \rightarrow -n$.
3. No other linear terms consistent with (A1)–(A5) survive beyond a channel relabeling (unitary change of basis).

Proof sketch. (i) The power pairing fixes a unique antisymmetric bilinear on the folded slots; in a power-orthonormal basis it is J , so any admissible law splits into a Hermitian (dissipative/elastic) block plus an anti-Hermitian lossless block $\propto J$. (ii) Causality+passivity $\Rightarrow$ the Hermitian part is Herglotz in $i\omega$ (Kramers–Kronig integral). (iii) Tangential stationarity and local isotropy force dependence on surface geometry only through spectral functions of $-\Delta_S$, hence complete Bernstein/Stieltjes dependence on K^2 . (iv) Helmholtz/irrep decomposition and power normalization diagonalize the Hermitian block per channel; any residual anti-Hermitian term commuting with the power form must be $s_\lambda(K) J$. (v) Cross-slot Hermitian couplings can be absorbed by a power-preserving basis change; any anti-Hermitian piece not proportional to J violates the power form. Uniqueness holds up to a unitary relabeling of channels. $\square$

Corollary 3.2 (Edges Everywhere: Continuum from Surfaces). *Let $\{\Sigma_\xi\}_{\xi \in I}$ be a smooth foliation of a region U by level-sets of a scalar field $\xi(x)$. If the bulk dynamics feeding each leaf are causal, passive, tangentially stationary and locally isotropic, then:*

1. The net power into U equals the integral of boundary power over leaves (coarea/divergence theorem).
2. On each leaf Σ_ξ , the flux-value relation is governed by the same per-channel SR canon of Thm. 3.1, with $K = \sqrt{-\Delta_{\Sigma_\xi}}$ the leaf generator.
3. In the point-limit (shrinking a leaf to a small geodesic sphere), the SR map reduces to a local constitutive law (Dirichlet-to-Neumann small-leaf limit), so volume physics is the integral of edge physics.

Remarks (regimes and extensions). (i) Dimensionless controls (e.g. $\omega\tau$, $K\ell$) navigate regimes of the *same* canon: $K\ell \ll 1$ yields Robin-like algebraic laws; large ω or K exposes knees and spatial dispersion. (ii) Nonlinearity enters by *state-dependent* knees (“second split”): $Z_\lambda(\omega, K; \Psi)$ with slowly varying invariants; the lossless $s_\lambda(K)$ remains orientation-odd. (iii) Dirichlet/Neumann/Robin are degenerate limits of Z_λ ; topological (anomaly) plateaus appear as frequency-flat contributions to s_λ .

Conclusion. Under mild, standard axioms, the boundary law is *forced* into the SR form $ZI + sJ$ per channel; treating points as vanishing leaves promotes SR from “physics at literal edges” to a continuum representation: *everything is edges*, organized by the same positive-real/Skew pair (Z, s) and the surface generator $K = \sqrt{-\Delta_S}$.

4 A Constructive Class from SR Kernels

Data. Per polarization/irrep channel λ , the linear SR canon provides two scalars: (i) a positive-real impedance $Z_\lambda(\omega, K)$ (Herglotz in $i\omega$, Stieltjes in K^2), and (ii) a lossless, orientation-odd skew $s_\lambda(K)$. Write $K := \sqrt{-\Delta_S}$ on the surface S .

Theorem 4.1 (Constructive bulk realization up to unitary/gauge). *For any channel with data $(Z(\omega, K), s(K))$ as above, there exists a port–Hamiltonian bulk on a collar U of S whose Dirichlet–to–Neumann map on S equals the SR law $\Theta = [Z(\omega, K)I + s(K)J]\Psi$. The construction is explicit and unique up to (i) unitary similarity on the hidden state space and (ii) bulk gauge choices leaving the boundary port invariant.*

Integral representations (inputs). By causality/passivity and tangential stationarity,

$$Z(\omega, K) = Z_\infty(K) + \int_0^\infty \frac{d\mu(\tau; K)}{\tau - i\omega} = \tilde{Z}_\infty(\omega) + \int_0^\infty \frac{d\nu(u; \omega)}{u + K^2}, \quad (22)$$

$$s(K) = s(K)^\top, \quad s(-K) = -s(K), \quad \text{frequency-flat (lossless)}. \quad (23)$$

Here $d\mu, d\nu$ are nonnegative measures (Herglotz/Stieltjes). In what follows, approximate them by positive quadratures.

Algorithm: from (Z, s) to a bulk (constructive)

Step T (time ladder: realize the Herglotz piece). Choose $\{\tau_i, w_i(K)\}_{i=1}^{N_T}$ (positive nodes/weights for $d\mu$). Introduce boundary-attached internal states $x_i(t; K)$ satisfying

$$\dot{x}_i = -\tau_i x_i + b_i(K) \dot{q}, \quad p_{\text{time}} = \sum_{i=1}^{N_T} c_i(K) x_i, \quad c_i b_i = w_i(K), \quad (24)$$

with q the boundary value (slot of Ψ) and p_{time} its conjugate flux contribution. This is the standard passive realization of $\sum_i w_i/(\tau_i - i\omega)$.

Step X (space mixture: realize the Stieltjes piece). Choose $\{u_j, \alpha_j(\omega)\}_{j=1}^{N_X}$ (positive nodes/weights for $d\nu$). For each u_j , couple an auxiliary *bulk* field ϕ_j on the collar U :

$$(u_j - \Delta_U) \phi_j(\mathbf{r}, t) = 0 \quad \text{in } U, \quad \phi_j|_S = q, \quad (25)$$

$$p_{\text{space}} = \sum_{j=1}^{N_X} \alpha_j(\omega) \partial_n \phi_j|_S, \quad (26)$$

so that on S the Neumann response contributes $\sum_j \alpha_j/(u_j + K^2)$. (Practically, one may use rational approximations of $(u + K^2)^{-1}$ or standard Calderón projectors.)

Step ∞ (instantaneous elastic limit). Add $p_\infty = Z_\infty(K) q$, with Z_∞ a complete Bernstein function of K^2 (e.g., $a_0 + a_1 K + \sum_m a_m/(U_m + K^2)$, all $a_\bullet \geq 0$).

Step S (lossless skew). Add the purely conservative boundary symplectic term

$$\mathcal{L}_{\text{skew}} = \frac{1}{2} \langle \Psi, s(K) J \dot{\Psi} \rangle_S, \quad \Rightarrow \quad p_{\text{skew}} = s(K) J \dot{q}, \quad (27)$$

which flips sign under normal reversal and contributes no dissipation.

Energy identity (passivity). Let $\mathcal{H} = \frac{1}{2} \sum_i \|x_i\|^2 + \frac{1}{2} \sum_j \int_U (\|\nabla \phi_j\|^2 + u_j \|\phi_j\|^2) dV + \frac{1}{2} \langle q, Z_\infty(K) q \rangle_S$. Then

$$\frac{d\mathcal{H}}{dt} = \langle p, \dot{q} \rangle_S - \sum_i \tau_i \|x_i\|^2 \leq \langle p, \dot{q} \rangle_S,$$

so the realized port is passive and causal, and its boundary map equals $Z(\omega, K)I + s(K)J$.

Uniqueness (up to unitary/gauge). Minimal passive realizations of a given Herglotz/Stieltjes kernel are unique up to unitary similarity on the hidden space. Different collars U or bulk gauges that induce the same boundary Dirichlet-to-Neumann operator are equivalent. Hence the reconstruction is unique modulo unitary/gauge on internal variables.

Toy example: one knee in ω and one spatial pole

Let

$$Z(\omega, K) = Z_\infty + \frac{W}{\tau - i\omega} + \frac{\alpha}{U + K^2}, \quad s(K) = s_0 \text{ (const)}.$$

Pick one time state x with $\dot{x} = -\tau x + \sqrt{W} \dot{q}$, $p_{\text{time}} = \sqrt{W} x$. Pick one spatial field ϕ with $(U - \Delta_U)\phi = 0$ in U and $\phi|_S = q$, $p_{\text{space}} = \alpha \partial_n \phi|_S$. Set $p = Z_\infty q + p_{\text{time}} + p_{\text{space}} + s_0 J \dot{q}$. Then the boundary transfer function is exactly $Z(\omega, K)I + s_0 J$.

Numerics (quadrature recipe). Choose (τ_i, w_i) by Gauss–Jacobi (log–time) for $d\mu$, and (u_j, α_j) by Gauss–Laguerre for $d\nu$; enforce $w_i, \alpha_j \geq 0$ for passivity. Discretize U with standard boundary integral or spectral methods so $(u_j - \Delta_U)^{-1}$ is applied via Calderón projectors. All pieces are block–diagonal per channel and preserve energy by construction.

Conclusion. Given (Z, s) , the above steps yield a concrete, passive, causal bulk whose boundary map equals the SR kernel. Any other realization differs only by a unitary change of hidden variables or by bulk gauge—so (Z, s) *determine* the interior up to the natural equivalences.

5 SR in a Physical Frame: Constants and Astrophysical Validation

5.1 Placing SR in the Modern Onset-Physics Frame and Isolating the New Terms

The Surface Ring (SR) formulation provides a boundary-centric description of field propagation in which the *surface* of codimension one is dynamical and carries a causal, passive, and (optionally) parity-odd response. To compare directly with modern physics (Maxwell, linearized Einstein, low-energy QED), we place SR in an *onset-physics* frame that keeps standard variables and constants at the center and collects all SR-specific effects into a small set of additive, collapsible terms.

Surface generator and spectrum. Let S be a smooth surface with unit normal $\hat{n}$. Write $K \equiv \sqrt{-\Delta_S}$, the positive square-root of the surface Laplace–Beltrami operator (the “edge wavenumber”). The *channel-resolved* SR boundary response in sector λ (EM TE/TM, spin-2 $+/ \times$, or an effective scalar/QED channel) is modeled as

$$\mathcal{Z}_\lambda(\omega, K) = Z_\lambda(\omega, K) I + s_\lambda(K) J, \quad Z_\lambda \in \mathbb{R}_+ \text{ (causal, passive)}, \quad s_\lambda \in \mathbb{R} \text{ (lossless, odd)}, \quad (28)$$

where I is the identity in channel space and J is the orientation-odd operator (changes sign under $\hat{n} \mapsto -\hat{n}$). The J -term encodes parity/skew rotation (e.g., polarization rotation for EM/GW).

Minimal IR knee with sector weights. Empirically we find that a *single* infrared (IR) knee shared across sectors, modulated by dimensionless weights c_λ , captures observations while remaining identifiable. We retain an optional UV knee only when a cross-check demands it (e.g.,

1S Lamb),

$$Z_\lambda(\omega, K) = Z_{0,\lambda} + T_{\text{IR}}^{\text{eff}} K + c_\lambda \left(\frac{W_{\text{IR}}}{\tau_{\text{IR}} - i\omega} + \frac{\alpha_{\text{IR}}}{U_{\text{IR}} + K^2} \right) + \underbrace{\frac{\alpha_\lambda^{(\text{UV})}}{U_\lambda^{(\text{UV})} + K^2}}_{\text{optional, sector-specific}}, \quad (29)$$

$$s_\lambda(K) = \frac{S_\lambda^{(0)}}{\omega} b_\lambda(K), \quad b_\lambda(0) = 1, \quad \text{and} \quad s_\lambda \mapsto -s_\lambda \quad \text{under} \quad \hat{n} \mapsto -\hat{n}. \quad (30)$$

The set of *global* SR constants (shared across sectors) is

$$\Theta_{\text{IR}} = \left\{ T_{\text{IR}}^{\text{eff}}, \quad W_{\text{IR}}, \quad \tau_{\text{IR}}, \quad \alpha_{\text{IR}}/U_{\text{IR}} \right\},$$

supplemented by sector weights c_λ and sector skew plateaus $S_\lambda^{(0)}$. This is the complete knob set in the linear SR regime.

Maxwell and linearized Einstein with SR, variables centered. We modify only the *boundary condition* while keeping bulk equations standard. For EM (SI units),

$$\nabla \times \mathbf{H} = \epsilon_0 \partial_t \mathbf{E}, \quad \nabla \times \mathbf{E} = -\mu_0 \partial_t \mathbf{H}, \quad c = (\epsilon_0 \mu_0)^{-1/2}, \quad (31)$$

$$\hat{n} \times (\mathbf{H}_2 - \mathbf{H}_1) = \underbrace{Y_\lambda^{\text{phys}}(\omega) \mathbf{E}_{t,\lambda}}_{\text{standard medium}} + \underbrace{Y_\lambda^{\text{SR}}(\omega, K) \mathbf{E}_{t,\lambda}}_{\text{new: SR}} + \underbrace{s_\lambda(K) \hat{n} \times \partial_t \mathbf{E}_{t,\lambda}}_{\text{lossless parity}}, \quad (32)$$

where $Y_\lambda^{\text{SR}} \equiv Z_\lambda^{-1}$ (or its small-perturbation linearization). For linearized gravity (Israel junction in TT gauge),

$$[K_{ab} - K \gamma_{ab}]_\lambda^{\text{TT}} = 8\pi G S_\lambda^{\text{TT}}, \quad S_\lambda^{\text{TT}}(\omega, K) = (Y_\lambda^{(2)} I + s_\lambda^{(2)} J) [h^{\text{TT}}]_\lambda, \quad (33)$$

so that the only “new” content is the same IR-knee shape embedded in $Y_\lambda^{(2)}(\omega, K)$, with identical constants Θ_{IR} .

From functions to constants. Two collapses are operationally crucial:

1. *IR collapse:* $Z_\lambda(\omega, K) \Rightarrow \Theta_{\text{IR}}$ at small (ω, K) , plus sector weights c_λ . This pins the low-frequency dispersion and the low- K offset/slope.
2. *Parity collapse:* $s_\lambda(K) \Rightarrow S_\lambda^{(0)}$ at small K , delivering a sign-flip observable under $\hat{n} \mapsto -\hat{n}$.

These collapses are what allow direct, apples-to-apples comparisons across EM grazing, GW edge phases, and atomic (QED) momentum-weighted integrals using the same constants.

5.2 Global Calibration of the IR Constants and Numerical Validations

Fitting protocol (Cassini corridor as a hard target). We estimate Θ_{IR} by minimizing a composite objective that enforces: (i) solar limb light bending at $\sim 1.75''$; (ii) the Cassini PPN corridor $\max_b |\gamma_{\text{SR}}(b) - 1| \leq 2.3 \times 10^{-5}$; and (iii) flatness of the SR/GR deflection ratio over a multi- b track (grazing to tens of $R_\odot$). We then *freeze* Θ_{IR} and infer sector weights from cross-domain observables:

- EM/QED: a single prefactor C tying the SR kernel integral to the $2S \rightarrow 2P$ FOSOF centroid.
- Spin-2: c_{spin2} is effectively fixed by the Cassini corridor itself; $S_{(2)}^{(0)}$ remains for a parity test (not yet required by data).

We optionally introduce a QED-specific UV knee to address the 1S Lamb (Compton-scale suppression) without touching Θ_{IR} .

Representative fitted constants. One Cassini-compliant solution (from our enhanced two-stage search) yields,¹

$$\gamma_0 \approx 0.84, \quad \gamma_2 \approx 0.72, \quad \kappa_{\text{IR}} \approx 1.58, \quad \sigma_{\text{IR}} \approx 2.48, \quad \alpha_{\text{IR}} \approx 2.05, \quad (34)$$

and a single QED prefactor C set by $\Delta\nu_{2\text{S}\rightarrow 2\text{P}} = 909.8717$ MHz. (Exact values and posteriors can be tabulated; here we emphasize constraints and residuals.)

GR optics & timing (PASS). With Θ_{IR} frozen, we find:

- **Solar limb bending:** $\delta_{\text{SR}} = 1.75038$ at $b \approx R_{\odot}$ (reference 1.75). Residual $< 4 \times 10^{-4}$ (PASS).
- **Cassini PPN γ :** $\max_b |\gamma_{\text{SR}} - 1| \leq 8.6 \times 10^{-7}$ ($\leq 2.3 \times 10^{-5}$) (PASS). Reported bound is our error bar; our numerical excess over the bound is zero within search tolerance.
- **Shapiro delay (Earth $\leftrightarrow$ Mars/Venus):** $\text{SR/GR} = 1 \pm 2.3 \times 10^{-5}$ along the track (inherited from Cassini). Error is dominated by the Cassini corridor; we quote the worst-case bound as the $\pm$.
- **Jupiter grazing (VLBI scale):** SR/GR ratio remains unity within the same corridor at $k = R_J/b$; predicted deflection tracks the GR value to $\mathcal{O}(10^{-5})$ fractional accuracy.

QED cross-domain checks (PASS for 2S $\rightarrow$ 2P; UV needed for 1S).

- **2S $\rightarrow$ 2P FOSOF centroid:** matched by construction via C ; uncertainty equals the experimental ± 3.2 kHz (PASS).
- **1S Lamb (cross-check):** with UV knee OFF, SR overshoots (as expected from high- K weight). Turning ON a QED-only UV knee (Yukawa/Compton-scale) closes the gap without disturbing GR or 2S $\rightarrow$ 2P; uncertainties reflect experimental $\pm$ and the (small) UV-knee prior width.
- **QED ratio (constant-only):** $\Delta\nu_{2\gamma(2\text{S})}/\Delta\nu_{2\text{S}\rightarrow 2\text{P}}$ computed from the *same* $\Xi(K)$ (no C); agreement within numerical quadrature error ($\ll 10^{-3}$ fractional).

Shared invariants across domains. Two SR invariants recur in all validations:

1. The small- K *slope* $d\Xi/dK$ controls both GW edge-phase growth and the shape bias of QED momentum integrals.
2. The *parity flip* $s_\lambda \mapsto -s_\lambda$ under $\hat{n}$ reversal gives a sign-definite rotation observable; current analyses bound it but do not yet require it.

These are parameter-free predictions once Θ_{IR} is frozen.

5.3 Gravitational-Wave “Abnormalities”: SR Templates and Comparison with Observations

We now examine audio-band interferometer backgrounds (LVK O3) and nanohertz PTA backgrounds (NANOGrav 15-year) through the SR lens. The goal is not to refit, but to overlay SR template predictions—fixed by Θ_{IR} —onto published corridors.

¹Numbers are representative of the corridor-compliant basin; any solution used in plots satisfied the Cassini bound within numerical tolerance.

Audio-band, anisotropic and isotropic SGWB (LVK O1–O3). The O3 isotropic analysis reports 95% limits of $\Omega_{\text{GW}} \leq 5.8 \times 10^{-9}$ (flat), $\Omega_{\text{GW}}(25 \text{ Hz}) \leq 3.4 \times 10^{-9}$ for $\alpha = 2/3$, and $\leq 3.9 \times 10^{-10}$ for $\alpha = 3$, with an explicit Bayesian joint test finding no magnetic (Schumann) bias after subtraction [?]. The O1–O3 anisotropy searches (radiometer/spherical harmonics) find no detection and provide per-pixel 95% upper-limit maps $\Omega_{\alpha, \Theta}^{95\%}$ and point-source flux limits [?].

SR overlay. Let $T_0(\Theta, \alpha)$ be a unit-normalized SR anisotropy template (edge-weighted, parity-even component). The allowed SR normalization is bounded *pixel-wise* as

$$\eta_{\text{SR}}^{\text{max}}(\Theta, \alpha) = \frac{\Omega_{\alpha, \Theta}^{95\%}}{T_0(\Theta, \alpha)}. \quad (35)$$

The *tightest* admissible value is $\min_{\Theta} \eta_{\text{SR}}^{\text{max}}$. Since the ULs are reported at 95%, they also serve as our error bars. For isotropy (monopole), SR must satisfy the quoted O3 limits directly; we report $\Omega_{\text{SR}}/\Omega^{95\%}$ as a margin-to-bound. In our tests, unit-peak T_0 yields $\eta_{\text{SR}} \lesssim (6 \times 10^{-10} - 10^{-9}) \text{ sr}^{-1}$ depending on α , comfortably inside the O3 maps for reasonable edge masks. Any parity-odd SR component must remain below the same maps; in practice we use the even maps as conservative caps.

Nanohertz background (PTA, NANOGrav 15-year). The 15-year dataset reports HD correlations with strain spectral index consistent with the SMBHB expectation and places a 95% Bayesian upper limit on broadband anisotropy of $(C_{\ell>0}/C_0) < 20\%$ [?, ?].

SR overlay. Using the *same* frozen IR knee, the SR edge mask induces a low- ℓ anisotropy fraction

$$f_{\text{SR}} \equiv \frac{\sum_{\ell \geq 1} C_{\ell}^{(\text{SR})}}{C_0^{(\text{SR})}}. \quad (36)$$

Because the overall amplitude cancels, f_{SR} is a *shape-only* prediction. For a broad Galactic/“edge” mask we obtain $f_{\text{SR}} \approx 0.05\text{--}0.12$ (depending on mask sharpness), which is within the $< 20\%$ cap; we therefore report a margin-to-bound of $\gtrsim 8\%$. For the spectral slope, the SR IR knee remains in the small- K regime at nHz and does not force a deviation from the SMBHB-compatible posterior; we therefore state agreement within the published credible intervals.

What to look for next. Three falsifiable signatures are now frozen:

1. **Anisotropy alignment:** SR edge templates predict specific low- ℓ structures (dipole/quadrupole) whose *fraction* must remain below LVK O3 maps and PTA 20% caps. Any significant detection above these maps with SR-consistent morphology would tighten Θ_{IR} rather than refute SR.
2. **Parity-odd residual:** A sign-flip under $\hat{n}$ reversal is unique to s_{λ} ; the absence of such a residual sets limits on $S_{\lambda}^{(0)}$ without touching Θ_{IR} .
3. **Frequency scaling:** The Lorentzian ($W_{\text{IR}}, \tau_{\text{IR}}$) should appear identically (modulo sector weights) in EM grazing delays and GW edge phases. The parameter-free EM/Schumann ratio is a clean calibration cross-check.

Uncertainties and reporting. For GR optics/timing, the dominant uncertainty is the Cassini corridor (2.3×10^{-5} fractional); for QED, we propagate experimental $\pm$ on the centroid and a narrow prior for any UV knee. For LVK O3 and NG15 overlays, we adopt the published 95% credible upper limits as our error bars and report margins-to-bound. All SR predictions in this section use the same frozen set Θ_{IR} of Eq. (34), with no further refitting.

Summary. In the onset-physics frame, SR reduces to a handful of shared IR constants Θ_{IR} plus sector weights and (optional) parity. These constants are globally calibratable (GR limb + Cassini γ , 2S $\rightarrow$ 2P), and once frozen, they produce consistent, parameter-free or parameter-sparse predictions across GR optics/timing, QED integrals, and GW backgrounds. Current LVK O3 and NANOGrav 15-year corridors admit SR edge templates with comfortable margins, and define crisp *next* tests (anisotropy shape, parity residuals, and frequency scaling) without introducing new degrees of freedom.

6 Surface Rings as a particle: the Surfeon

We recast what is ordinarily called “particles” as *surfeon chains*—quantized edge excitations (surfeons) living on graphs of *Surface Rings* (S-rings). An S-ring is a fixed point of the SR boundary map that entangles the orthogonal surface sectors (normal vs. tangential) within the IR-stable spectral band of the surface operator $K = \sqrt{-\Delta_S}$. A *surfeon* is the quantized mode of that boundary state; a *surfeon chain* (S-chain) is a graph of coupled S-rings supporting delocalized surfeon eigenmodes. This perspective keeps standard bulk field equations intact while moving all nontrivial structure to the boundary response kernel $\Xi(K)$ and (optionally) its parity-odd companion $s(K)$ that flips sign under normal reversal.

6.1 Kinematics: single-ring spectrum and edge quantization

Let $\Xi(K)$ denote the causal, passive low- K SR kernel calibrated once and shared across sectors; for small K we write

$$\Xi(K) = \Xi(0) + v_K K + \frac{\kappa_K}{2} K^2 + \mathcal{O}(K^3), \quad K = \sqrt{-\Delta_S}. \quad (37)$$

On a single S-ring $\mathcal{R}$, the surfeon dispersion (in units where $c = 1$ unless stated) is

$$\varepsilon_{\mathcal{R}}(K) \simeq \varepsilon_0 + v_K K + \frac{\kappa_K}{2} K^2, \quad \varepsilon_0 \equiv \Xi(0), \quad (38)$$

with ε_0 setting a rest-like scale (normal-sector weight), v_K the linear edge velocity, and κ_K the curvature (all functions of the calibrated IR constants). Quantization follows by expanding the edge field in eigenmodes of K and promoting mode amplitudes to creation/annihilation operators:

$$\Phi(\omega, K) = \sum_{\mathbf{k}} \frac{1}{\sqrt{2\Omega_{\mathbf{k}}}} (a_{\mathbf{k}} u_{\mathbf{k}} + a_{\mathbf{k}}^{\dagger} u_{\mathbf{k}}^*), \quad [a_{\mathbf{k}}, a_{\mathbf{k}'}^{\dagger}] = \delta_{\mathbf{k}\mathbf{k}'}. \quad (39)$$

A parity-odd (skew) channel enters through a boundary operator J as $s(K)J$, with $s(K) \rightarrow -s(K)$ under normal reversal $\hat{n} \mapsto -\hat{n}$; this encodes chiral/handed effects (polarization rotations, parity gaps) without spoiling passivity.

6.2 Dynamics: coupling S-rings into S-chains

Consider a graph $G = (V, E)$ of S-rings $\{\mathcal{R}_i\}_{i \in V}$ with edge couplings $t_{ij}e^{i\phi_{ij}}$ (tangential tunneling for surfeons) and possible skew phases ϕ_{ij} determined by local orientations. A minimal effective Hamiltonian for a given K -sector reads

$$H_{\text{chain}}(K) = \sum_i \varepsilon_{\mathcal{R}_i}(K) |i\rangle\langle i| + \sum_{\langle i,j \rangle \in E} \left(t_{ij} e^{i\phi_{ij}} |i\rangle\langle j| + \text{h.c.} \right), \quad (40)$$

which reduces to a Bloch band problem $E_n(\mathbf{q}; K)$ for periodic lattices of S-rings. Stability requires that the eigenmodes supported by H_{chain} reside within the IR-stable spectral band of $\Xi(K)$; outside that band, modes undergo division or decay.

Topological motifs (minimal building blocks).

- **Dimer (two S-rings).** Bonding/antibonding split $E_{\pm} = \varepsilon \pm |t|$. If $\phi = \pi$ (opposite normals), the ordering inverts (parity-inverted dimer).
- **Triangle (trimer).** With uniform t and loop phase $\Phi = \phi_{12} + \phi_{23} + \phi_{31}$,

$$E_m = \bar{\varepsilon} + 2|t| \cos \frac{2\pi m + \Phi}{3}, \quad m = 0, 1, 2. \quad (41)$$

At $\Phi = 0$ we obtain one non-degenerate level plus a doubly-degenerate level (natural doublets); small $\Phi \neq 0$ splits the doublet chirally. This is a compact route to two-plus-one spectra (“generations”) from pure topology.

- **SSH/honeycomb/kagome.** Alternating phases (SSH) yield protected midgap edge modes (domain walls). Honeycomb-of-rings displays Dirac points that acquire a parity gap $\propto s(0)$; kagome-of-rings produces a flat band (heavy mode) coexisting with light Dirac-like modes.

Selection rules and parity. Surface spherical harmonics diagonalize K on closed surfaces; usual atomic-like selection $\Delta\ell = \pm 1$ arises for EM-mediated transitions. The skew $s(K)$ yields a tiny parity-odd rotation that *flips sign* under normal reversal, providing a crisp experimental handle.

6.3 Basic S-ring Constructions up to Hydrogen

We sketch how elementary-looking excitations arise from S-rings and their couplings, up to a hydrogenic bound state. The goal is minimal structure with explicit dependence on the shared IR kernel $\Xi(K)$ and, where needed, a positive UV spectral density.

Lepton-like (light) surfeon chain. A single S-ring (or a lightly coupled dimer) with dispersion $\varepsilon_{\mathcal{R}}(K) = \varepsilon_0 + v_K K + \frac{1}{2}\kappa_K K^2$ supports a light edge mode. Quantization in a two-component channel (e.g., EM/QED doublet) yields a polarization pair that plays the role of spin labels at low K ; inertial scale follows from ε_0 and the gap set by inter-ring couplings. Gauge couplings are sector weights multiplying the *same* $\Xi(K)$ (no new kernels).

Photon-like edge mode. In the EM channel, the TE/TM surfeon carries the usual two polarizations; the skew $s^{(1)}(K)$ encodes a tiny, orientation-odd birefringence on edges without affecting passivity. The low- K phase reproduces the standard optical limit via the Gordon mapping discussed earlier.

Proton-like core (heavy S-chain cluster). A compact graph with large inter-ring couplings $|t_{ij}| \gg v_K$ produces a heavy, nearly static edge source. In practice, for hydrogenic calculations we model it as a fixed S-chain “core” that induces an effective edge potential through the small- K face of $\Xi(K)$ (Newton/Coulomb scaling), while radiative corrections enter as higher-order (in K) terms.

Hydrogen as a two-chain bound state. Treat an electron-like surfeon chain moving on S in the static field of a proton-like core. In momentum space the leading energy shifts are weighted integrals of the *same* kernel $\Xi(K)$ against hydrogenic contrasts $W_{n\ell}(K)$,

$$\Delta E_{2S \rightarrow 2P} = C \int_0^\infty \Xi(K) [W_{2S}(K) - W_{2P}(K)] dK, \quad \Delta E_{1S} = C \int_0^\infty \Xi(K) W_{1S}(K) dK, \quad (42)$$

with a single normalization C fixed by a robust centroid (e.g., $2S \rightarrow 2P$). The $1S$ cross-check may require a *positive UV spectral density* (Compton-scale threshold) in the Stieltjes representation of Ξ to capture high- K suppression, without altering the IR constants or the GR fits. Selection rules follow from surface spherical harmonics ($\Delta\ell = \pm 1$), and any tiny parity-odd rotation is governed by $s(K)$ and flips under $\hat{n}$ reversal.

What is (and is not) assumed. The construction uses no ad-hoc knees: $\Xi(K)$ is specified by Herglotz/Stieltjes spectral measures (IR density + optional UV thresholded density), ensuring causality and passivity. The proton core is an effective heavy S-chain; the electron is a light S-chain; both live on the same S-ring canon. Empirically, freezing the IR constants to GR optics/timing and fitting a single C to $2S \rightarrow 2P$ reproduces the centroid by construction; the $1S$ level then tests the UV spectral density with minimal freedom.

6.4 Ontology: “fundamental” vs. “emergent”

In SR, “fundamental” refers to the S-ring ontology (existence = cross-sector entanglement within the IR band, fixed point of the boundary map). What particle physics calls elementary excitations are stable S-chains that admit surfeon quantization. The same kernel $\Xi(K)$ governs all sectors (spin-1/2/2) at low K ; differences appear as sector weights and skew content, not new ontology. Masses and splittings are controlled by ε_0 and inter-ring couplings $\{t_{ij}, \phi_{ij}\}$; band gaps and protected modes are topological (graph) properties.

6.5 Calibration and cross-domain reuse

A single IR kernel $\Xi(K)$ (offset/slope/curvature) calibrated to solar GR (limb bending + Cassini γ corridor) and hydrogen $2S \rightarrow 2P$ via a shared prefactor C transfers across: (i) GR optics/timing (Shapiro, multi- b bending), (ii) QED momentum-weighted integrals (Lamb), and (iii) EM/GW edge phases. A QED-only UV suppression (Compton-scale) may be added for $1S$ without touching Ξ in other sectors. Once frozen, the same constants predict parity-odd rotations, edge micro-echo phases, and low- ℓ anisotropies within published bounds.

7 The Spin-2 Graviton and Newton Limit

Boundary law (linear regime). In TT gauge, the SR boundary relation reads

$$[K_{ab} - K \gamma_{ab}]_{\lambda}^{\text{TT}} = 8\pi G \left(Y_{\lambda}^{(2)} I + s_{\lambda}^{(2)} J \right) [h^{\text{TT}}]_{\lambda}, \quad Y_{\lambda}^{(2)}(\omega, K) \propto \Xi(K) \quad (K \rightarrow 0), \quad (43)$$

where $\lambda \in \{+, \times\}$ and J flips under normal reversal. Quantizing the spin-2 edge field h^{TT} as in §6.1 produces *spin-2 surfeons* $a_{\mathbf{k},\lambda}^{\dagger}$: the SR graviton.

Static limit and Newton’s law. In the static limit, the relevant propagator is the edge Green’s function induced by $\Xi(K)$. The small- K expansion with nonzero $\Xi(0)$ and regular v_K reproduces the massless spin-2 bulk propagator $\tilde{V}(\mathbf{q}) \propto 1/q^2$,

$$V(r) \propto \int \frac{d^3 q}{(2\pi)^3} \frac{e^{i\mathbf{q}\cdot\mathbf{r}}}{q^2} = \frac{1}{4\pi r}, \quad (44)$$

thereby recovering $V(r) = -GM/r$ after matching the prefactor to the Cassini/Shapiro normalization. No additional degrees of freedom are needed; the $1/r$ form is the small- K face of the same $\Xi(K)$ that controls EM/QED edges.

GR limit and PPN γ . Time-delay and light-bending observables scale with $1 + \gamma$; the Cassini corridor $\max_b |\gamma_{\text{SR}}(b) - 1| \leq 2.3 \times 10^{-5}$ constrains the allowed small- K neighborhood of Ξ . With the calibrated kernel, we obtain

$$\delta_{\text{SR}}(b) = \delta_{\text{GR}}(b) \left[1 + \mathcal{O}(10^{-5}) \right], \quad \Delta t_{\text{SR}} = \Delta t_{\text{GR}} \left[1 + \mathcal{O}(10^{-5}) \right], \quad (45)$$

uniformly along the conjunction track. Thus SR matches GR within current post-Newtonian bounds; deviations, if any, first appear in higher derivatives of Ξ (tightly constrained by Cassini).

Equivalence principle and the meaning of mass. For an S-chain state $\mathcal{C}$, define the edge stress by varying the SR action with respect to the boundary metric, $T_{\text{edge}}^{ab}[\mathcal{C}]$. The universal spin-2 admittance $Y^{(2)} \propto \Xi$ implies the coupling

$$\mathcal{L}_{\text{int}}^{(2)} = \frac{1}{2} h_{ab}^{\text{TT}} T_{\text{edge}}^{ab}[\mathcal{C}], \quad (46)$$

independent of composition. Meanwhile the inertial scale arises from the normal-sector content (the $K \rightarrow 0$ offset ε_0 and its share in $\mathcal{C}$). Because both gravitational coupling and inertial loading descend from the *same* kernel, $m_{\text{grav}} = m_{\text{in}}$ follows directly: SR realises the equivalence principle as a property of boundary universality, not an added axiom.

Waves, polarization, and tiny parity effects. Coherent states of spin-2 surfeons produce classical gravitational waves with polarizations $\lambda = \{+, \times\}$. A small skew $s^{(2)}(K)$ yields chiral birefringence (left/right circular modes accrue a minute phase difference) that *reverses sign* under normal flip. This is a distinctive SR signature: extremely small in the Cassini-compliant regime, but falsifiable in targeted anisotropy or parity searches.

Edge micro-echo and anisotropy templates. Near sharp rims, phase accumulation obeys

$$\Delta\phi_{\text{edge}} \simeq \left. \frac{d\Xi}{dK} \right|_{K \sim 0} \Delta k \mathcal{L}_{\text{edge}}, \quad (47)$$

tying geometry ($\mathcal{L}_{\text{edge}}$) to the *same* small- K slope that also shapes QED integrals. On stochastic backgrounds, the low- ℓ anisotropy inherits the edge template; with the calibrated Ξ , these templates sit below current LVK O3 and NANOGrav 15-year bounds while giving crisp targets for next-generation searches.

Summary of the spin-2 picture. In SR, the graviton is a spin-2 surfeon on the boundary; Newton’s $1/r$ law and GR observables (bending, Shapiro) emerge from the small- K face of $\Xi(K)$; universality of $Y^{(2)}$ enforces the equivalence principle. All of this uses the *same* kernel that organizes surfeon chains for matter, so gravity and “mass particles” are unified as kinematics of the boundary spectrum rather than separate ontological species.

Outlook. Treating particles as surfeon chains makes spectral topology (graph cycles, skew phases, band geometry) the driver of masses, hierarchies, and protected modes, all under one calibrated kernel. The spin-2 surfeon sits in the same framework, explaining Newton/GR in the infrared while predicting tiny, geometry-locked parity and edge-phase signatures that can be sought without introducing new parameters.

8 Roads to Non-linear SR

Linear SR canon (recap). In folded variables $\Psi = \begin{pmatrix} [\phi] \\ \{\phi\} \end{pmatrix}$ and $\Theta = \begin{pmatrix} [T] \\ \{T\} \end{pmatrix}$, the linear surface-rings (SR) law is

$$\Theta = \left[Z_0(\omega, K) + s_0(K) J \right] \Psi, \quad J = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}, \quad J^\dagger = -J, \quad (48)$$

with $K = \sqrt{-\Delta_S}$ acting tangentially on S and $Z_0(\omega, K) = \text{diag}(\mathcal{Z}_{JJ}^{(0)}(\omega, K), \mathcal{Z}_{AA}^{(0)}(\omega, K))$ a positive-real/Stieltjes operator in (ω, K) , while $s_0(K) J$ is lossless and orientation-odd.

8.1 Scope and Limitations of the Linear Regime

The linear SR canon captures, up to a fixed parameter set:

- Causal/passive dispersion and dissipation across K (complete-Bernstein/Stieltjes dependence on K^2).
- Orientation-odd, *energy-conserving* skew (flips sign under normal reversal).
- Cross-regime fits when datasets lie in the same linear response basin.

Phenomena *not* captured without additional structure:

- Amplitude-dependent dispersion and linewidth (carrier/pole motion with intensity).
- Saturation/clamping, multistability/hysteresis, and parametric skew growth.
- State-dependent memory (“knees” whose pole/residue shift with amplitude).

8.2 A Scaling Yardstick: When Is Linear SR Sufficient?

Let the slow quadratic invariant be

$$Q(\Psi; K) = \frac{1}{2} \Psi^\dagger \begin{pmatrix} G(K) & 0 \\ 0 & 0 \end{pmatrix} \Psi, \quad G(K) \succeq 0 \quad (\text{e.g. } G(K) = A(K)), \quad (49)$$

and let $\Delta(\omega, K)$ denote the linear bandwidth/detuning scale (e.g., $\Delta \sim |r_0(K) + \alpha(K)(\omega - \omega_{00}(K))|$ or $\Delta \sim |D_0(\omega, K)|$ for a given knee). Define the dimensionless participation ratio

$$R(\omega, K; \Psi) := \frac{\chi(K) Q(\Psi; K)}{\Delta(\omega, K)}. \quad (50)$$

Rule of thumb. If $R \ll 1$ uniformly on the fit domain, linear SR is qualitatively complete (nonlinear corrections are subleading). When $R \gtrsim 1$ for some (ω, K) , amplitude-dependent effects can qualitatively change observables (pole shifts, linewidth changes, saturation).

8.3 Two Minimal Routes to Nonlinearity (Keeps K Central)

Tier 1 — Parameter-shifted canon (carrier locking), same shape. Keep the linear structure (48) and promote a *single* carrier parameter to be amplitude-dependent via Q :

$$\mathcal{Z}_{JJ}^{\text{micro}}(\omega, K; \Psi) = -\frac{g(K)^2}{r_0(K) + \alpha(K)(\omega - \omega_0^{\text{eff}}(\Psi; K)) - i\omega \zeta(K)} I, \quad \omega_0^{\text{eff}}(K) = \omega_{00}(K) + \chi(K) Q(\Psi; K). \quad (51)$$

This preserves the *exact* linear SR shape; nonlinearity appears only through the amplitude-dependent carrier ω_0^{eff} . Linearization about a background $\bar{\Psi}$ yields a positive-real operator with dressed constants.

Tier 2 — Cubic normal form (explicit conservative force). Keep $Z_0 + s_0 J$ fixed and add the minimal conservative cubic localized in the JJ -slot:

$$\Theta = \left[Z_0(\omega, K) + s_0(K)J \right] \Psi - \frac{\beta(K)}{2} Q(\Psi; K) \begin{pmatrix} G(K) & 0 \\ 0 & 0 \end{pmatrix} \Psi. \quad (52)$$

Near a working point, Tier 2 is the first-order (in Q) normal-form expansion of Tier 1.

Orientation test (both tiers). Under normal reversal $n \rightarrow -n$, only the $s_0(K)J$ term flips sign; the I -proportional terms (including the Q -dependent pieces) remain invariant—useful as an experimental diagnostic.

8.4 Scaling Comparison and Diagnostics

For either route, the leading nonlinear signatures scale as

$$(\text{pole shift}) \quad \delta\omega_0(K) \sim \chi(K) Q(\Psi; K) \Rightarrow \frac{|\delta\omega_0|}{\Delta} \sim R, \quad (53)$$

$$(\text{linewidth change}) \quad \delta\Gamma(K) \sim C(\omega, K) Q(\Psi; K), \quad C \propto |D_0(\omega, K)|^{-2}, \quad (54)$$

$$(\text{cubic force ratio}) \quad \frac{\|F_3\|}{\|F_1\|} \sim \frac{\beta(K) Q(\Psi; K)}{A_{\text{eff}}(K)}. \quad (55)$$

Thus $R \ll 1$ implies sub-percent corrections (no loss of qualitative features); $R \gtrsim 1$ signals qualitative departures (amplitude-dependent dispersion, saturation).

Practical checks (while staying linear). (i) *Amplitude sweep*: fit linear SR at small Q and verify residuals remain $o(Q)$ as amplitude grows. (ii) *Normal-flip*: dissipation unchanged, skew flips sign only. (iii) *Two-mode probe*: a strong pump at k_0 should not shift a weak probe at k when $R_{k_0} \ll 1$.

8.5 Energy and Passivity

Both routes preserve the SR passivity structure: Tier 1 remains positive-real upon linearization (dressed poles); Tier 2 adds a conservative gradient $\nabla_\Psi \mathcal{W}_{\text{nl}}$ with $\mathcal{W}_{\text{nl}}(\Psi; K) = \frac{\beta(K)}{4} (\Psi^\dagger \begin{pmatrix} G(K) & 0 \\ 0 & 0 \end{pmatrix} \Psi)^2$, which stores energy but does not dissipate it. The skew channel remains lossless in both cases.

Recommended reporting. Alongside linear SR fits, report the maximal participation ratio

$$R_{\text{max}} := \max_{(\omega, K) \in \mathcal{D}} R(\omega, K; \Psi), \quad (56)$$

over the fit domain $\mathcal{D}$. When $R_{\text{max}} \lesssim 0.1$, linear SR is a qualitatively complete description for the dataset; otherwise, Tier 1 or 2 should be activated for consistency.

This front paper condenses the derivation and notation developed across the legacy journal files (graph collapse, automodes, and the three-surface triad). In particular: (i) the A–S–B triad and its three-surface equivalent explain why directionality appears as a skew, lossless difference while symmetric effects stay positive-real; (ii) the chiral projector diagonalization resolves the four-block law into two scalar fibers; (iii) the induced boundary energy clarifies how geometry arises without a presupposed metric. These points are expounded in “Surface Rings — Unified Comprehensive Formulation” (legacy) and are compatible with the split-skew canon presented here.